

THAI RECIPE AGENT

IBM Watsonx Orchestrate | Agentic AI Project | RAG-Based

Domain: Generative AI • Retrieval-Augmented Generation • Culinary Assistance

PRESENTED BY



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Powered by

**IBM Watsonx
Orchestrate**

Agentic AI • RAG Retrieval • Conversational Intelligence

Problem Statement

THE CHALLENGE

Mr. Ramesh owns a premium Thai restaurant that serves unique, exclusive dishes. The recipes are created in-house and are not publicly available. These proprietary recipes represent the restaurant's identity and competitive advantage.

Currently the recipes are managed manually, making it difficult to:

- Preserve recipe knowledge systematically
- Train new chefs efficiently
- Generate variations of dishes
- Provide intelligent recipe assistance
- Maintain consistency in taste and preparation

THE OBJECTIVE

Develop a Retrieval-Augmented Generation (RAG) based Thai Recipe Generator using Generative AI to design and build an AI-powered system that can:

- Store and retrieve proprietary Thai recipes securely
- Generate recommendations based on ingredients, cuisine type, spice level or customer preferences
- Answer chef queries on preparation, ingredients, techniques and serving styles
- Suggest new recipe variations while maintaining authentic Thai flavors
- Help train restaurant staff using conversational AI

Proposed Solution

A RAG-powered Thai Recipe Agent, built on IBM Watsonx Orchestrate, that transforms the restaurant's proprietary recipe archive into an intelligent, conversational culinary assistant.



Secure Knowledge Vault

Ingest proprietary recipe documents into a private knowledge base — accessible only to the authorised agent.



RAG Retrieval Engine

Retrieves the most relevant recipe passages before the LLM generates, ensuring authentic and accurate answers.



Conversational Chef Assistant

Step-by-step ingredients, quantities and techniques delivered through a natural chat interface.



Serving-Scaled Quantities

Automatically adjusts ingredient amounts when the user specifies a different number of servings.



Grounded, Hallucination-Free

Answers strictly from the provided knowledge base — no external internet look-ups.



Staff Training Companion

Helps train new chefs and kitchen staff with consistent, on-demand guidance.

Technology Used

0

IBM Cloud

1

Enterprise-grade cloud platform hosting the full agentic AI workload.

0

IBM Watsonx Orchestrate

2

Agent build & orchestration platform used to create, test and deploy the recipe agent.

0

RAG Architecture

3

Retrieval-Augmented Generation pipeline — retrieves recipe context before generating a response.

0

GPT-OSS 120B (OpenAI)

4

Large language model powering natural, chef-friendly recipe explanations.

0

Knowledge Ingestion

5

Uploaded proprietary Thai recipe document indexed as the sole knowledge source.

0

Watsonx Agent Channels

6

Native chat surface used to interact with the agent in real time.

Orchestrate Components Used

1

Profile

Defines the agent's identity, purpose and public description.

2

Knowledge

Uploaded Thai recipe document — the sole grounded source for responses.

3

Quick-Start Prompts

Ready-made prompts (e.g. Massaman Curry, Tom Yum) for one-tap access.

4

Behavior

System instructions that shape tone, scope and response style.

5

Channels (Preview)

Interactive chat surface used to test and deploy the agent.

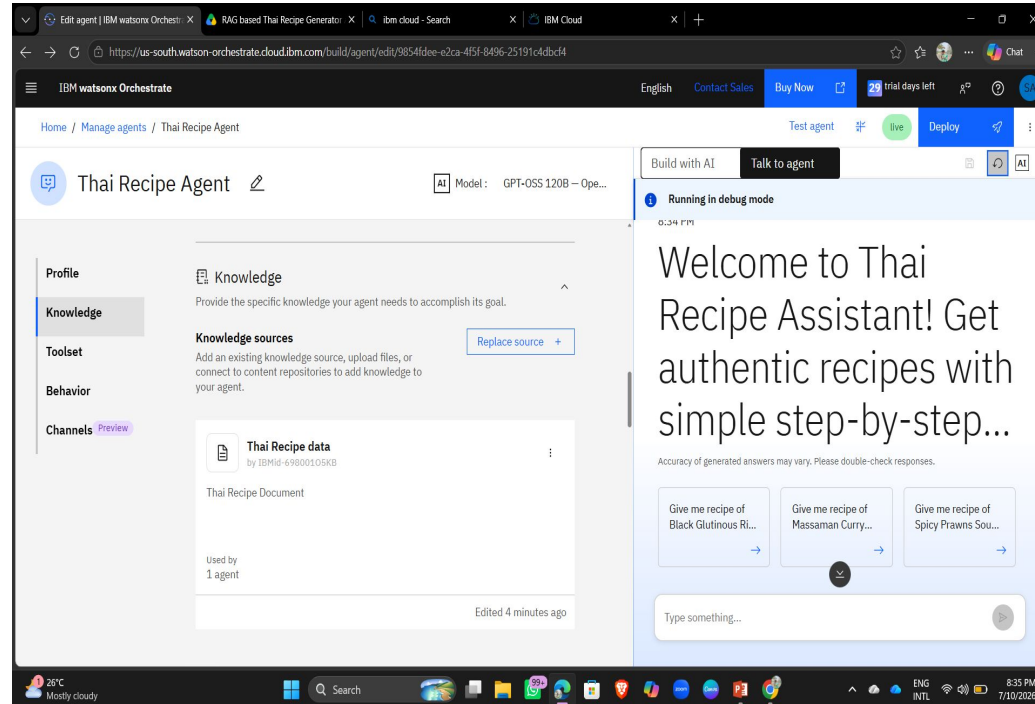
Agent Overview & Profile

The screenshot displays the IBM Watsonx Orchestrate web interface. The top navigation bar includes links for 'Edit agent', 'RAG based Thai Recipe Generator', 'ibm cloud - Search', and 'IBM Cloud'. The main header shows 'IBM watsonx Orchestrate' with options for 'English', 'Contact Sales', 'Buy Now', and a '29 trial days left' indicator. The breadcrumb trail is 'Home / Manage agents / Thai Recipe Agent'.

The left sidebar contains a menu with 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels' (marked as 'Preview'). The 'Profile' section is active, showing a 'Legal Notice' link and a 'View all' button. The 'Profile' description area contains the text: 'The Objective of this project is to build an AI-powered Thai Recipe Assistant that helps users prepare authentic Thai dishes easily and confidently'. Below this, there is a prompt: 'Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.' and a link 'What makes a good description?'. A notification banner at the top of the profile section reads 'Notifications: 1 Dismissed | 1 total'.

The right panel shows the 'Talk to agent' interface. It features a 'Build with AI' button and a 'Talk to agent' button. Below these, a status bar indicates 'Running in debug mode'. The main chat area displays a welcome message: 'Welcome to Thai Recipe Assistant! Get authentic recipes with simple step-by-step...'. A disclaimer states: 'Accuracy of generated answers may vary. Please double-check responses.' Below the disclaimer are three buttons: 'Give me recipe of Black Glutinous Ri...', 'Give me recipe of Massaman Curry...', and 'Give me recipe of Spicy Prawns Sou...'. At the bottom, there is a text input field with the placeholder 'Type something...' and a 'Send' button.

Knowledge-Based RAG



Knowledge Source Name

The Thai Recipe Document

Created By

IBMId — 69800105KB

Coverage

Authentic Thai dishes — ingredients, quantities, step-by-step preparation, chef tips and serving styles.

Role in the Agent

The single grounded source used by the RAG pipeline. Every response is retrieved from and cited to this document.

Agent Behaviour & Instructions

The screenshot displays the IBM Watsonx Orchestrate interface for configuring a 'Thai Recipe Agent'. The interface is divided into several sections:

- Header:** Includes navigation links like 'Home / Manage agents / Thai Recipe Agent', 'Test agent', 'live', and 'Deploy'. It also shows a '29 trial days left' notification.
- Agent Profile:** Shows the agent name 'Thai Recipe Agent' and the model 'GPT-OSS 120B - Ope...'. It includes tabs for 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels'.
- Behavior Section:** Expanded to show instructions for the agent. The instructions define the agent's role as a 'Thai Recipe Assistant' and provide specific rules for handling requests, such as providing clear, simple, and beginner-friendly instructions, and keeping responses short and practical.
- Live Chat Interface:** On the right, a preview of the agent's chat interface is shown. It displays a welcome message: 'Welcome to Thai Recipe Assistant! Get authentic recipes with simple step-by-step...'. Below the message are three sample prompts: 'Give me recipe of Black Glutinous Ri...', 'Give me recipe of Massaman Curry...', and 'Give me recipe of Spicy Prawns Sou...'. A text input field at the bottom is labeled 'Type something...'.

System Architecture



End-to-End Flow

- ▶ User types a request (e.g. "Give me the recipe of Tom Yum Goong for 2 servings").
- ▶ The Watsonx Orchestrate agent applies its Behavior rules — Thai food scope only, knowledge base only.
- ▶ The RAG retriever fetches the matching recipe passages from the uploaded Thai Recipe Document.
- ▶ The LLM composes a structured answer: ingredients table, step-by-step method and quick tips.
- ▶ If the user requests a different serving size, quantities are scaled automatically.
- ▶ The final response is streamed back to the user through the Orchestrate chat channel.

Agent Output — 1 / 6

The user requests a Spicy Prawns Soup (Tom Yum Goong) recipe — the agent begins with the ingredients list for one serving.

The screenshot displays the IBM Watsonx Orchestrate web interface. The left sidebar shows the configuration menu with 'Behavior' selected. The main area shows the 'Thai Recipe Agent' configuration, including its profile, knowledge base, and behavior rules. The right panel shows the agent's output in 'Running in debug mode'.

Behavior Configuration:

- Profile:** Define how and where your agent should react to requests and respond to users.
- Knowledge:** Define the knowledge base for the agent.
- Toolset:** Define the tools available to the agent.
- Behavior:** Define the rules for the agent's behavior.
- Channels:** Define the channels through which the agent can be accessed.

Behavior Rules:

- You are a Thai Recipe Assistant.
- Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only.
- Rules:
 - Answer only Thai food and recipe related questions.
 - Use only the provided knowledge base. Do not use internet or external knowledge.
 - Provide clear, simple, and beginner-friendly instructions.
 - Explain recipes step-by-step so anyone can easily cook the dish.
 - By default, provide ingredient quantities provided in the document (one serving).
 - If the user specifies the number of servings, adjust ingredient quantities accordingly.
 - Include ingredients, cooking steps, and simple tips when available.
 - Keep responses short, practical, and easy to understand.
 - If a recipe is not available in the knowledge base, politely say you do not have that information.

Agent Output:

You 8:48 PM
Give me recipe of Spicy Prawns Soup with Chili Paste

Thai Recipe Agent 8:48 PM | Show Reasoning
Spicy Prawns Soup with Chili Paste (Tom Yum Goong)
Serves 1 (adjust by multiplying the amounts)

Item	Amount
Thai prawns (peeled)	3
Chicken or vegetable stock	300 g

Save as test

Type something...

Agent Output — 2 / 6

Continued ingredient list — straw mushrooms, lemongrass, coriander roots, fish sauce, lime juice, kaffir lime leaves and bird's-eye chillies.

The screenshot displays the IBM Watsonx Orchestrator web interface. The top navigation bar includes links for 'Home', 'Manage agents', and 'Thai Recipe Agent'. The main content area is divided into two panels. The left panel, titled 'Thai Recipe Agent', contains a sidebar with 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels'. The 'Behavior' section is active, showing instructions for the agent. The right panel, titled 'Running in debug mode', displays a list of ingredients and their quantities, along with a chat input field.

Behavior Instructions:

Define how and where your agent should react to requests and respond to users.

Instructions

"You are a Thai Recipe Assistant.
Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only.
Rules:
- Answer only Thai food and recipe related questions.
- Use only the provided knowledge base. Do not use internet or external knowledge.
- Provide clear, simple, and beginner-friendly instructions.
- Explain recipes step-by-step so anyone can easily cook the dish.
- By default, provide ingredient quantities provided in the document (one serving).
- If the user specifies the number of servings, adjust ingredient quantities accordingly.
- Include ingredients, cooking steps, and simple tips when available.
- Keep responses short, practical, and easy to understand.
- If a recipe is not available in the knowledge base, politely say you do not have that information.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Running in debug mode

Chicken or vegetable stock	300 g
Straw mushrooms (sliced)	100 g
Lemongrass, lightly crushed & cut 1-in pieces	100 g
Coriander roots (cleaned)	2 g
Fish sauce	2 Tbsp
Lime juice	2 Tbsp
Kaffir lime leaves, shredded	3 leaves
Bird's-eye chillies, lightly crushed	1-2
Each coriander leaves (for garnish)	5 leaves

Save as test

Type something...

Agent Output — 3 / 6

Final ingredients (chilli paste, evaporated milk) followed by the start of the preparation steps.

The screenshot displays the IBM Watsonx Orchestrator web interface. The top navigation bar includes links for 'Edit agent', 'RAG based Thai Recipe Generator', and 'IBM Cloud'. The main header shows 'IBM watsonx Orchestrator' with options for 'English', 'Contact Sales', 'Buy Now', and a '28 trial days left' indicator. The breadcrumb trail is 'Home / Manage agents / Thai Recipe Agent'. The left sidebar contains a menu with 'Profile', 'Knowledge', 'Toolset', 'Behavior' (selected), and 'Channels'. The 'Behavior' section is active, showing a 'Define how and where your agent should react to requests and respond to users.' area. The 'Instructions' field contains the following text:

"You are a Thai Recipe Assistant.
Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only.
Rules:
- Answer only Thai food and recipe related questions.
- Use only the provided knowledge base. Do not use internet or external knowledge.
- Provide clear, simple, and beginner-friendly instructions.
- Explain recipes step-by-step so anyone can easily cook the dish.
- By default, provide ingredient quantities provided in the document (one serving).
- If the user specifies the number of servings, adjust ingredient quantities accordingly.
- Include ingredients, cooking steps, and simple tips when available.
- Keep responses short, practical, and easy to understand.
- If a recipe is not available in the knowledge base, politely say you do not have that information.

The right panel shows the 'Running in debug mode' output. It lists ingredients and their quantities:

Bird's-eye chilies, lightly crushed	1-2
Fresh coriander leaves (for garnish)	5 leaves
Chili paste (optional, for extra heat)	1 Tbsp
Evaporated milk (optional, for a creamy version)	100 g

Below the ingredients, the 'Steps' section is visible, starting with '1. Prepare the aromatics' and '2. Lightly crush the lemongrass and rinse the coriander roots.' The third step is '3. Boil the stock'. A chat input field at the bottom right contains the text 'Type something...' and a 'Save as test' button.

Agent Output — 4 / 6

Preparation steps — preparing aromatics, boiling the stock, straining and adding mushrooms.

The screenshot displays the IBM Watsonx Orchestrate web interface. The top navigation bar includes links for 'Home', 'Manage agents', and 'Thai Recipe Agent'. The main content area is divided into two panels. The left panel, titled 'Thai Recipe Agent', shows the 'Behavior' tab selected in the left-hand menu. The 'Behavior' section contains a 'Profile' field with the text 'You are a Thai Recipe Assistant. Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only.' and a 'Rules' section with a list of instructions. The right panel, titled 'Running in debug mode', shows a list of steps for the recipe: 1. Prepare the aromatics, 2. Lightly crush the lemongrass and rinse the coriander roots, 3. Boil the stock, 4. In a pot, bring the chicken/vegetable stock to a boil, 5. Add the lemongrass and coriander roots. Boil a few minutes to infuse flavor, 6. Strain (optional), 7. If you prefer a clear broth, pour the stock through a sieve, discarding the solids. Return the clear broth to the pot, 8. Add mushrooms, 9. Bring the broth back to a boil and add the straw mushrooms. At the bottom of the right panel, there is a 'Type something...' input field and a 'Save as test' button.

IBM watsonx Orchestrate

Home / Manage agents / Thai Recipe Agent

Test agent live Deploy

Build with AI Talk to agent

1 Running in debug mode

1. Prepare the aromatics
 - 2. Lightly crush the lemongrass and rinse the coriander roots.
3. Boil the stock
 - 4. In a pot, bring the chicken/vegetable stock to a boil.
 - 5. Add the lemongrass and coriander roots. Boil a few minutes to infuse flavor.
6. Strain (optional)
 - 7. If you prefer a clear broth, pour the stock through a sieve, discarding the solids. Return the clear broth to the pot.
8. Add mushrooms
 - 9. Bring the broth back to a boil and add the straw mushrooms.

Save as test

Type something...

Agent Output — 5 / 6

Cooking stage — adding the prawns, seasoning and finishing the soup with acidity and heat.

The screenshot displays the IBM Watsonx Orchestrate web interface. The top navigation bar includes links for 'Edit agent', 'RAG based Thai Recipe Generator', and 'IBM Cloud'. The main header shows 'IBM watsonx Orchestrate' with options for 'English', 'Contact Sales', 'Buy Now', and a '28 trial days left' indicator. The breadcrumb trail is 'Home / Manage agents / Thai Recipe Agent'. The left sidebar contains a menu with 'Profile', 'Knowledge', 'Toolset', 'Behavior' (selected), and 'Channels'. The main content area is titled 'Thai Recipe Agent' and shows the 'Behavior' configuration. The 'Instructions' section contains the following text: 'You are a Thai Recipe Assistant. Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only. Rules: - Answer only Thai food and recipe related questions. - Use only the provided knowledge base. Do not use internet or external knowledge. - Provide clear, simple, and beginner-friendly instructions. - Explain recipes step-by-step so anyone can easily cook the dish. - By default, provide ingredient quantities provided in the document (one serving). - If the user specifies the number of servings, adjust ingredient quantities accordingly. - Include ingredients, cooking steps, and simple tips when available. - Keep responses short, practical, and easy to understand. - If a recipe is not available in the knowledge base, politely say you do not have that information.' Below the instructions, there is a text input field with the placeholder 'Specify what your agent should do, how it should respond and any restrictions it should follow.' On the right side, the 'Running in debug mode' panel shows a list of steps: 10. Cook the prawns, 11. Add the peeled prawns, 12. Season, 13. Stir in the fish sauce and shredded kaffir lime leaves, 14. Finish with acidity and heat, 15. Remove the pot from heat, 16. Tip: Lime juice is most aromatic when added after the soup is off the heat, and 17. Optional creamy twist. At the bottom of the debug panel, there is a 'Save as test' button and a text input field with the placeholder 'Type something...'. The bottom status bar shows the system clock as 8:51 PM on 7/10/2026, along with weather and network icons.

Agent Output — 6 / 6

Final serve step and Quick Tips from the agent on fish sauce balance and adding lime juice off-heat.

The screenshot displays the IBM Watsonx Orchestrate web interface. The top navigation bar includes links for 'Edit agent', 'RAG based Thai Recipe Generator', and 'IBM Cloud'. The main header shows 'IBM watsonx Orchestrate' with options for 'English', 'Contact Sales', 'Buy Now', and 'trial days left'. The breadcrumb trail indicates 'Home / Manage agents / Thai Recipe Agent'. The left sidebar contains a menu with 'Profile', 'Knowledge', 'Toolset', 'Behavior' (selected), and 'Channels'. The main content area is titled 'Thai Recipe Agent' and shows the 'Behavior' configuration. The 'Instructions' section contains the following text: 'You are a Thai Recipe Assistant. Your job is to help users cook authentic Thai dishes using information from the provided knowledge base only. Rules: - Answer only Thai food and recipe related questions. - Use only the provided knowledge base. Do not use internet or external knowledge. - Provide clear, simple, and beginner-friendly instructions. - Explain recipes step-by-step so anyone can easily cook the dish. - By default, provide ingredient quantities provided in the document(one serving). - If the user specifies the number of servings, adjust ingredient quantities accordingly. - Include ingredients, cooking steps, and simple tips when available. - Keep responses short, practical, and easy to understand. - If a recipe is not available in the knowledge base, politely say you do not have that information.' The right sidebar shows the 'Running in debug mode' section with a 'Serve' step and a 'Quick Tips' section. The 'Quick Tips' section includes the following text: 'Fish sauce only becomes aromatic when added to the boiling soup. Lime juice should be added after removing the pot from heat for the brightest flavor. Adjust the number of chilies or add extra chili paste to suit your spice tolerance. Enjoy your authentic Tom Yum Goong!'. Below the 'Quick Tips' section is a 'Save as test' button and a text input field with the placeholder 'Type something...'. The bottom status bar shows the system clock as 8:51 PM on 7/10/2026.

IBM watsonx Orchestrate

Home / Manage agents / Thai Recipe Agent

Thai Recipe Agent

Model: GPT-OSS 120B — Ope...

Running in debug mode

19. Serve

- 20. Ladle the soup into a bowl. Garnish with fresh coriander leaves.

Quick Tips

- Fish sauce only becomes aromatic when added to the boiling soup.
- Lime juice should be added after removing the pot from heat for the brightest flavor.
- Adjust the number of chilies or add extra chili paste to suit your spice tolerance.

Enjoy your authentic Tom Yum Goong!

Save as test

Type something...

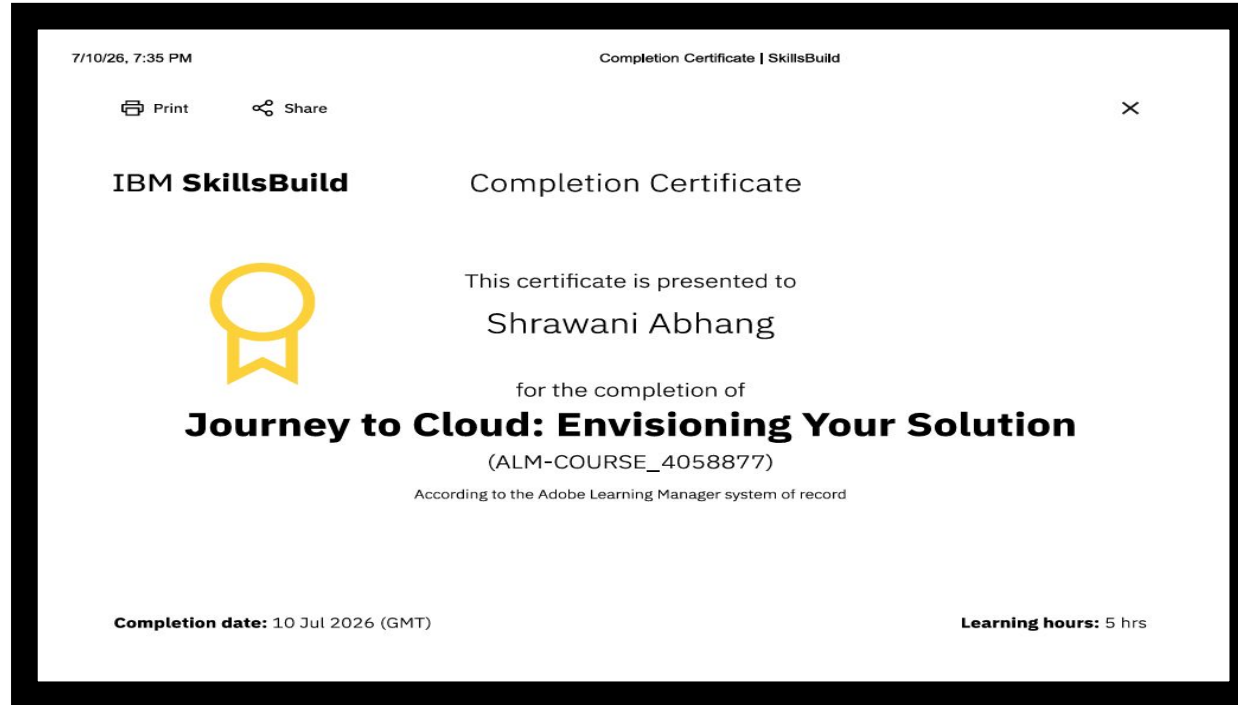
Certificates Earned:-



Certificates Earned:-



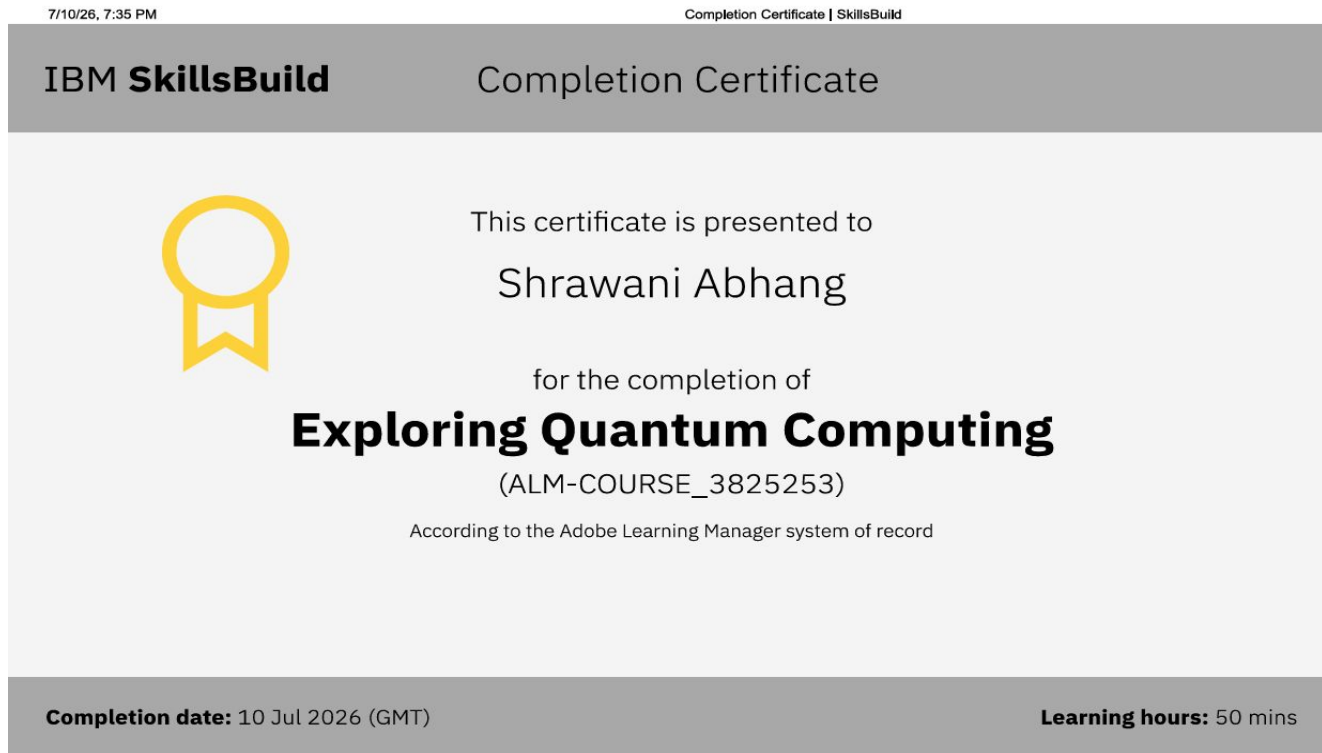
Certificates Earned:-



Certificates Earned:-



Certificates Earned:-



Certificates Earned:-



THANK YOU

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